ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Temperature and its Measurement** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** fever	Above 37 °C.
2	**(a)** because of the kink	Kink holds mercury.
3	**(d)** 30 °C	Room temperature.
4	**(c)** below 35 °C	So the reading is accurate.
5	**(b)** bulb is sensitive to body heat	Hand heat changes the reading.
6	**(a)** 40 °C	Between the two.
7	**(b)** iron	Iron conducts heat quickly to hand.
8	**(d)** water evaporating from the skin takes heat away	Evaporation cools.
9	**(b)** the metal conducts heat to the hand faster	Rate of conduction.
10	**(a)** surroundings	Heat flows from higher temperature.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Above normal 37 °C (1); The person has a fever (1).	1+1
12	Air trapped between layers (1); Air insulates (1).	1+1
13	Keeping tea hot (1); Keeping cold drinks cool (1).	1+1
14	Bottom is hotter than the liquid (1); It would give a wrong reading (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heat flows to room (1); Water cools (1); Becomes about 25 °C (1).	1+1+1
16	Shiny walls reflect heat back into the flask (1); Less heat escapes (1); Keeps drink hot or cold (1).	1+1+1
17	Allows heat to escape (1); Absorbs sweat (1); Keeps cool (1).	1+1+1
	OR: Heat one end of each (1); Place wax pins (1); Compare which falls first (1).	1+1+1
18	Iron conducts heat quickly (1); Wood conducts poorly (1); Rate differs (1).	1+1+1
	OR: Iron gives heat faster (1); Wood gives slowly (1); Rate of heat transfer differs (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rise = 8 °C each (1); After 2 min: 38 (1); After 4 min: 46 (1); After 10 min = 30 + 5 x 8 (1); 70 °C (1).	1+1+1+1+1
	OR: Hot water at 70 °C, cold water at 20 °C (1); Heat flows from hot to cold (1); Final in between (1); No more flow once equal (1); Same temperature (1).	1+1+1+1+1
20	Mercury expands (1); Rises in the narrow tube (1); Kink prevents falling (1); Heating would break it (1); Range is limited (1).	1+1+1+1+1
	OR: Mercury can leak (1); Glass may break (1); It has no kink (1); Not for body (1); Use clinical or digital (1).	1+1+1+1+1

Mistakes to avoid:

- Using a clinical thermometer for hot water or for a room; it is only for the **human body**.
- Not shaking the clinical thermometer before use, so an old reading is mistaken for the new one.
- Letting the bulb of the laboratory thermometer touch the beaker, or reading it from a slant.
- Believing that our hands can measure temperature accurately; touch can be **misleading**.
- Writing that cold flows from ice to a hand; **heat** flows from the hotter to the colder object.
- Writing the wrong units or values: body temperature is about 37 degrees C, not 100 degrees C.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Temperature and its Measurement** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** it can break	High temperature breaks it.
2	**(a)** mass	It measures temperature only.
3	**(c)** fall	Heat goes from spoon to glass.
4	**(b)** air	Air is a poor conductor.
5	**(a)** faster at first then slower	The difference decreases.
6	**(b)** copper rod	Copper is an excellent conductor.
7	**(b)** trapped air reduces heat loss	Trapped air.
8	**(c)** normal	About normal.
9	**(c)** it expands uniformly and does not wet glass	Mercury properties.
10	**(d)** air layer between them insulates	Trapped air.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The mercury needs time to reach the temperature of the liquid (1); An early reading may be wrong (1).	1+1
12	It conducts and spreads heat evenly (1); It is thin and light (1).	1+1
13	Heat flows from tea to surroundings (1); Surroundings gain heat (1).	1+1
14	Direct sunlight heats the bulb (1); So reading is of air only (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Water seeps through pores (1); Evaporation takes heat (1); So water cools (1).	1+1+1
16	Mercury takes about a minute (1); Digital beeps in seconds (1); Digital is safer (1).	1+1+1
17	Below 0 °C ice (1); Between 0 and 100 °C liquid (1); Above 100 °C steam (1).	1+1+1
	OR: Fever is rising (1); Doctor must be consulted (1); Monitor regularly (1).	1+1+1
18	Mercury would expand (1); Thermometer may break (1); Use cold or lukewarm (1).	1+1+1
	OR: Heat conducts upward (1); Metal conducts (1); Plastic spoon would stay cool (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pour hot water in can (1); Place on a table (1); Measure every few minutes (1); Temperature falls (1); Tends to room temperature (1).	1+1+1+1+1
	OR: Place each spoon in hot water (1); Same depth (1); Touch the free ends after 2 min (1); Iron is hottest (1); Wood and plastic remain cool (1).	1+1+1+1+1
20	Saucepan: aluminium (1); Handle: plastic or wood (1); Glove: thick cotton (1); Kettle body: steel (1); Choice by conduction (1).	1+1+1+1+1
	OR: Metal conducts heat to the water (1); Water heats quickly (1); Plastic is a poor conductor of heat (1); The handle stays cool (1); It is safe to hold (1).	1+1+1+1+1

Mistakes to avoid:

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- Letting the bulb of the laboratory thermometer touch the beaker, or reading it from a slant.
- Believing that our hands can measure temperature accurately; touch can be **misleading**.
- Writing that cold flows from ice to a hand; **heat** flows from the hotter to the colder object.
- Writing the wrong units or values: body temperature is about 37 degrees C, not 100 degrees C.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Temperature and its Measurement** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** see a doctor immediately	High fever.
2	**(b)** it is designed for body range	Range limit.
3	**(c)** to trap air	Reduces heat loss.
4	**(c)** below and above 0 °C	Range of a fridge.
5	**(a)** laboratory thermometer	Range up to 110 °C.
6	**(d)** across a metal rod	Conductors.
7	**(a)** 100 °C	Fixed points.
8	**(a)** melting ice	Ice-water mixture.
9	**(a)** plastic	Insulator.
10	**(b)** temperature of water falls	Ice absorbs heat.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	For even heating (1); Metal conducts heat well (1).	1+1
12	Metal conducts heat from the hand quickly (1); Hand loses heat so it feels cold (1).	1+1
13	Hot object cools (1); Water warms up (1).	1+1
14	Take readings with a thermometer (1); Compare the values (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It needs time to reach the same temperature (1); Early reading is incorrect (1); Steady reading is reliable (1).	1+1+1
16	Dip in melting ice: should read 0 °C (1); Dip in boiling water: should read 100 °C (1); Compare (1).	1+1+1
17	Snow traps air (1); Air is a poor conductor (1); Body heat stays (1).	1+1+1
	OR: Conducts heat from flame (1); Spreads heat evenly (1); Cooks quickly (1).	1+1+1
18	Touch is unreliable (1); A thermometer gives the exact temperature (1); A doctor can then decide the treatment (1).	1+1+1
	OR: Not to burn the hand (1); Poor conductor (1); Safe use (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Weather: forecasts (1); Health: fever (1); Cooking: right temperature (1); Storage: fridge (1); Industry: control (1).	1+1+1+1+1
	OR: They trap air (1); Air is an insulator (1); Heat loss reduces (1); Body or tea stays warm (1); Principle: insulation (1).	1+1+1+1+1
20	Take milk in a cup (1); Hold the thermometer vertically (1); Do not touch the bottom (1); Wait (1); Read level (1).	1+1+1+1+1
	OR: Double glass wall (1); Vacuum in between (1); Cork stopper (1); No medium for conduction (1); Outer plastic case (1).	1+1+1+1+1

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